

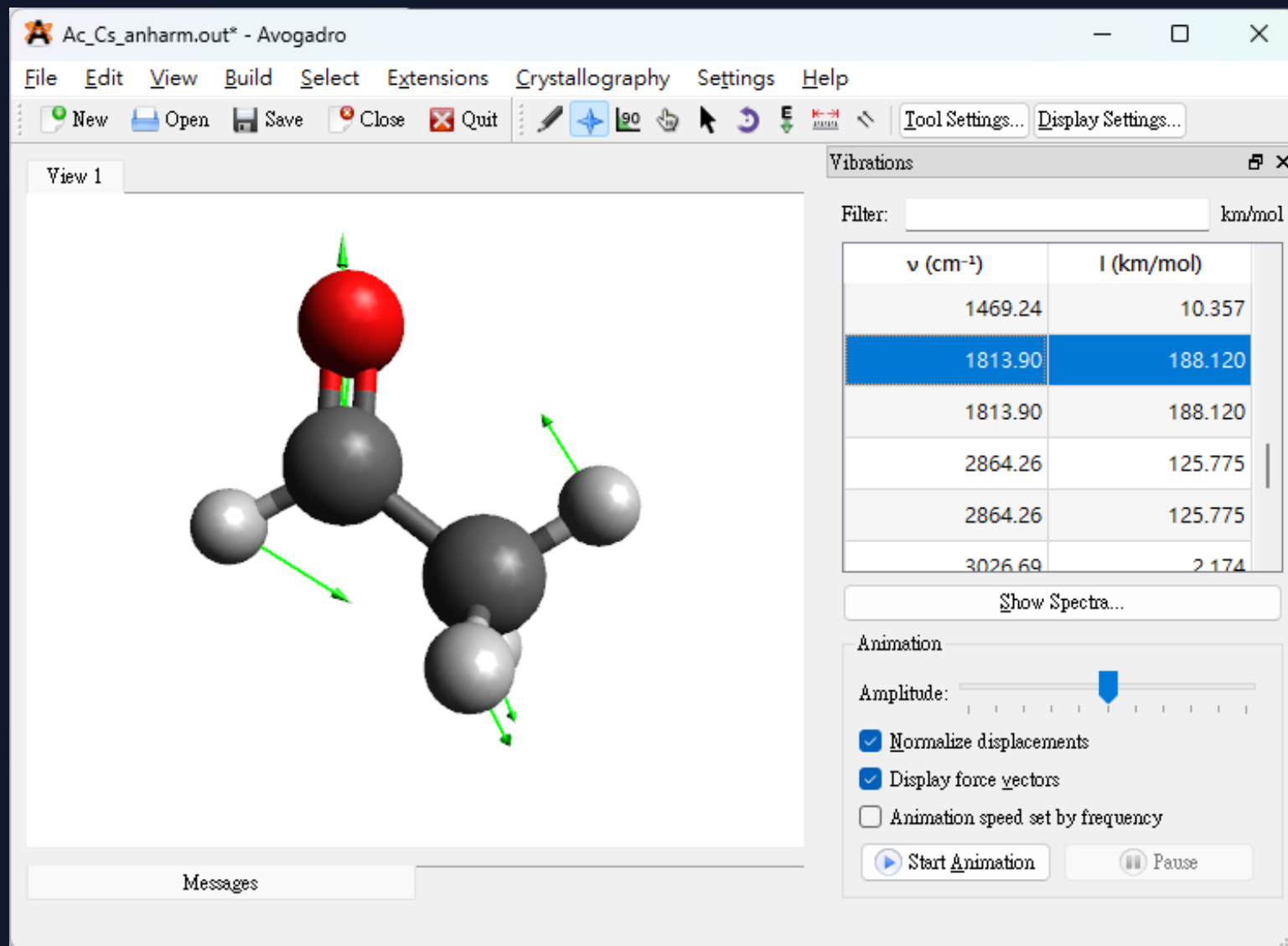
Building a Custom Vibrational Mode Viewer with **Superpowered** Claude Code

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Vibrational analysis NEEDS visualizer, but...

- We perform **anharmonic vibrational analysis**, and we need Visualizer to see how the mode motion looks like.
- Avogadro crashes when opening a second molecule while a mode animation is playing
- GaussView costs \$1,150 per additional platform
 - Mac users in the group can't use it



vib_viewer — a browser-based toolbox

Vibration Analyzer Spectra Viewer

Click or drop .log / .out / .hess / .fchk / .json / .cjson

Ac_Cs_anharm.out

7 atoms, 30 modes loaded.

	Index	Freq
#25	1469.2 cm ⁻¹	m=1.05 I=10.4
#26	1813.9 cm ⁻¹	m=9.30 I=188.1
#27	2864.3 cm ⁻¹	m=1.08 I=125.8
#28	3026.7 cm ⁻¹	m=1.04 I=2.2
#29	3078.7 cm ⁻¹	m=1.10 I=7.0
#30	3140.0 cm ⁻¹	m=1.10 I=9.5

Anim Quiver Both

Amplitude 1.0

Speed 1.0

Arrow len 2.2

Appearance

Export

Advanced

The 3D model shows a molecule with green arrows indicating atomic displacements for the selected vibrational mode.

- Single HTML file — open multiple windows simultaneously, no crash
- Loads Gaussian & ORCA output directly
- Animate modes, export GIF for slides
- Hosted on GitHub Pages — collaborators use it without HPC login
- Alternative layout for mobile devices

*First working version finished in 30 min.
Built in a few evenings.
Still used daily.*

Claude Code + Superpowers

Claude Code

- Terminal-based agentic coding tool
- Reads, writes, and runs files directly in your project
- You've probably heard of it by now

+

Superpowers plugin

- 236k stars on Github, extremely popular
- 14 skills covering the full dev cycle
- **Questions first** — scientists know what they want
- Spec → Plan → Build
- Parallel subagents for independent tasks
- Systematic debugging
- Verify before moving on

Github: <https://github.com/obra/Superpowers>

Claude Code: `/plugin install superpowers@claude-plugins-official`

Codex: available via the official Codex plugin marketplace

Before a single line of code was written

Brainstorming – Ask questions before code

Question	Answer
Animation only, or also quiver vectors?	Both
Cartesian or internal coordinate displacements?	Cartesian for now
How should the mode list be displayed?	Scrollable, sort by index or
Single amplitude slider or per-mode?	Single global slider
Rendering style?	Ball-and-stick, adjustable size
Color scheme?	CPK colors (like Avogadro)
Should the user be able to edit bonds?	Yes, with apply/restore
File formats to support?	Modified Gaussian log — example provided

~20 questions total.

Scientists are used to this — it's just experimental design before the experiment.

What came out of a 5-minute Q & A

Vibration Mode Viewer — Design Spec

Date: 2026-06-11

Status: Approved

Overview

A single self-contained HTML file (no build step, open directly in browser) that loads a molecular vibrational mode dataset and visualizes atomic displacements in 3D using Three.js. Supports Gaussian log files (standard and modified MBH/local mode format) and a canonical JSON format.

Architecture

Single HTML file with inline JavaScript organized into named sections:

1. Element data table (CPK colors + van der Waals radii, all 118 elements, Jmol scheme)
2. File parsers (Gaussian log auto-detect, JSON pass-through)
3. Three.js scene setup
4. Molecule rendering (ball-and-stick)
5. Arrow rendering (quiver)
6. Animation loop
7. UI / event handlers

Planned future refactor: split into `index.html` + `parser.js` + `renderer.js` + `ui.js` (Approach C) once multiple parsers accumulate. No structural changes needed to enable this — the sections map directly to files.

Data Schema

Internal representation (also the accepted JSON file format):

- 221-line design spec
 - what the tool should do
- 1413-line implementation plan
 - 12 tasks, each with verification steps and exact code snippets
- Learn a lot from the spec / plan
 - a map of the project



Without vs. With Superpowers



Same goal, same starting file, same model (Sonnet 4.6), same person.

	Default Claude Code	+Superpowers
First action	768 lines of code	Ask 20 clarifying questions
Planning artifacts	No planning phase	Spec → 1413-line plan
Time to first working version	~15 min	~20 min
Token usage	~15% of session	~30% of session
CPK colors, atom sizing	Required correction turns	(Asked) Correct by turn 1
3D displacement arrows	Required correction turns	(Asked) Correct by turn 1
IR spectrum click-to-select	Not added	Added unprompted
GIF recorder	Working, needed 1 fix	Working, needed 1 fix

Superpowers costs more upfront — in time and tokens.

7 *You get it back in fewer correction turns and spec / plan you can actually read.*

It keeps growing because we own it

- IR spectrum panel with Lorentzian broadening
- Animated GIF export
- Hessian analyzer —local mode / MBH definition
- Spectra viewer — overlay multiple spectra in browser
 - Show assignment in calculated spectra
- Remote file browser — load files from HPC directly
- Embedded / presentation mode
- Dark / light theme

92

Commits

since June 11, 2026

What I found

1 Superpowers benefits the user as much as the output.

The planning phase teaches me: how the problem can be decomposed, what decisions matter, and why certain approaches were chosen.

2 Our domain knowledge is our prompting superpower.

"Make it look like Avogadro" works better than "improve the visualization."
We already know what good looks like.

3 Planning-first resonates with how scientists think.

We design experiments before running them. controls, failures, and interpretations.
Superpowers enforce AI does the same for code.

4 This is where AI helped me most — not in my main research.

There are always tasks - not our main research questions but still consume time and attentions. AI helps me reduce the friction, so I get more time for my research.

Try it

Live tool

GitHub Pages



https://appendko.github.io/vib_viewer/

Source code

GitHub



https://github.com/Appendko/vib_viewer

Full workflow

HackMD



<https://hackmd.io/@Append/HJAJYGoZMg>

Questions?

Hessian Analyzer

From manual bookkeeping to click-and-see

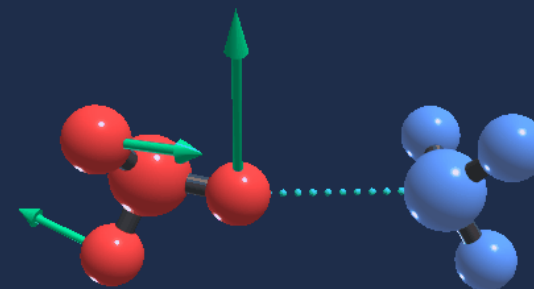
Before:

- Open visualizer, group atoms by eye
- Manually write group definitions
 - e.g. `[[1,3,4],[2,5]]`
- Run the code on HPC
- Download output and visualize
 - check if the grouping made sense
 - Repeat if wrong



After:

- Load the Hessian file into the viewer
 - Remote access to HPC now
- Click atoms to assign groups
- See the resulting mode immediately



Spectra Viewer

Compare spectra without leaving the browser

Before:

- Python/matplotlib - fast but static
 - no interactive assignment
- Some users export to Origin for overlay, scaling, and offset
- Assignment: which transition is this peak?
 - Go back to the output file, check manually



After:

- Drag calculated and experimental spectra into the browser — anyone can do it
- Overlay immediately, adjust scaling, offset, and broadening on the fly
- Drag and Drop for top-bottom subfigures
- Click any peak → assignment shown live
 - no need to go back to the output file
- Export PNG when everything looks right

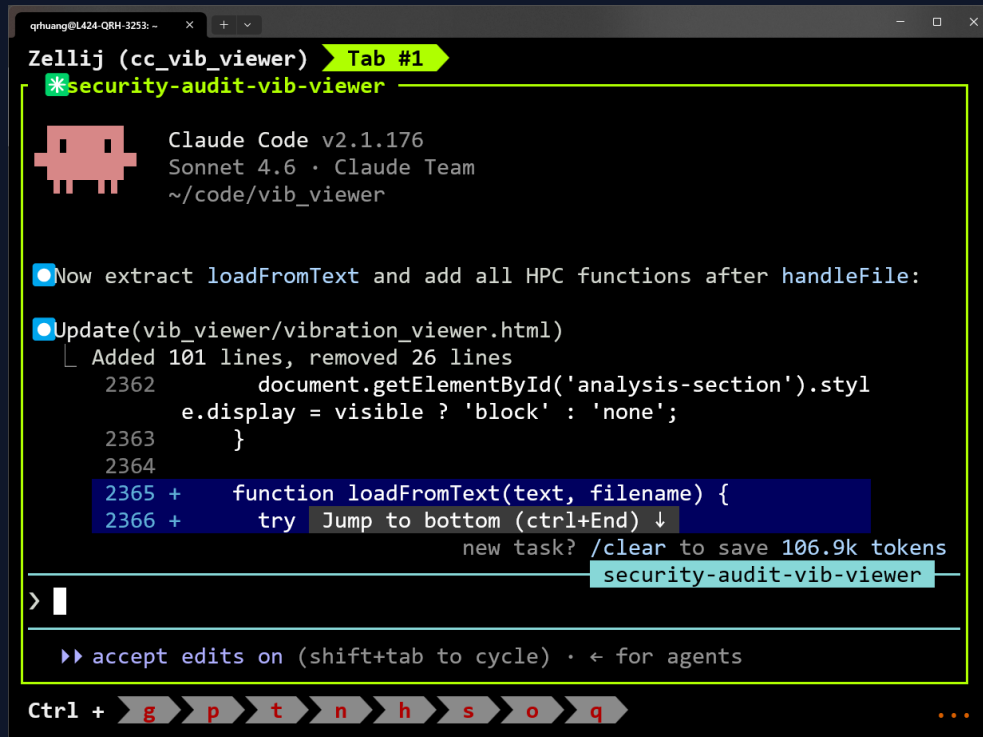
Spectra Viewer



Working from anywhere — Zellij + Tailscale

Working from anywhere without losing your session

Claude Code runs in a terminal - if the connection drops mid-session, context is lost
I have 5 machines: Mac Mini (running Claude Code), 2 desktops, 2 laptops, ...



```
Zellij (cc_vib_viewer) Tab #1
*security-audit-vib-viewer

Claude Code v2.1.176
Sonnet 4.6 · Claude Team
~/code/vib_viewer

Now extract loadFromText and add all HPC functions after handleFile:

Update(vib_viewer/vibration_viewer.html)
└─ Added 101 lines, removed 26 lines
2362     document.getElementById('analysis-section').styl
      e.display = visible ? 'block' : 'none';
2363   }
2364
2365 +   function loadFromText(text, filename) {
2366 +     try Jump to bottom (ctrl+End) ↓
                                new task? /clear to save 106.9k tokens
security-audit-vib-viewer
> |

accept edits on (shift+tab to cycle) · ← for agents

Ctrl + g p t n h s o q
```

Zellij

terminal multiplexer

Keeps the session
alive even when you
close the terminal or
switch machines

+

Tailscale

zero-config VPN

Connects all your
devices into a private
network, no port
forwarding needed

14 *Start a session on any machine, walk away, continue from another - as if I never left my desk.*